

## # SIH26151 " Final Edge-Case and Risk Register

**\*\*Purpose:\*\*** Mandatory implementation and test register. A branch cannot be accepted if its applicable P0 edge cases are not tested.

### ## Priority definitions

- **\*\*P0:\*\*** Required for SIH demo and before using any non-fixture input.
- **\*\*P1:\*\*** Required before a live authorized pilot.
- **\*\*P2:\*\*** Required for mature operational deployment.

### ## Edge-case register

| ID    | Priority | Condition                                    | Required control                                                               | Verification                                        |
|-------|----------|----------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------|
| EC-01 | P0       | Source timeout/offline/moved/503             | Status event, retain history, bounded retries                                  | Fixture transitions offline -> changed              |
| EC-02 | P0       | Mirrors/reposts/duplicate jobs               | Content hash + independence group + idempotency                                | Same content twice counts once                      |
| EC-03 | P1       | Malicious/oversized/malformed/binary content | Quarantine, MIME/size/CPU limits, no JS                                        | Hostile fixture cannot crash parser                 |
| EC-04 | P0       | Login/CAPTCHA/JS-only/form-based source      | Record not-collected reason; never bypass                                      | Fixture creates safe access limitation              |
| EC-05 | P1       | Source authorization expires                 | Approval/expiry/kill switch/blocklist                                          | Expired source job blocked                          |
| EC-06 | P1       | Tor proxy loss/direct-egress risk            | SOCKS5h-only, direct egress deny, health check                                 | Collector cannot access direct test endpoint        |
| EC-07 | P0       | OnionScan timeout/crash/schema change        | Pinned artifact, isolated timeout, raw result/error artifact, fixture fallback | Invalid output does not break pipeline              |
| EC-08 | P0       | Shared/stale cert/header/CDN                 | Rarity/freshness/caveat/cap                                                    | Infra-only cannot reach likely tier                 |
| EC-09 | P0       | Mixer/escrow/shared wallet                   | Wallet role/downweight/cap                                                     | Shared-wallet fixture stays weak                    |
| EC-10 | P0       | Published/revoked/shared/fake PGP            | Verify state/role; distinguish signature verification                          | Published-key fixture weaker than signature fixture |
| EC-11 | P0       | Recycled/confusable alias                    | Original+normalized storage, collision warning/cap                             | Unicode/common alias test no merge                  |
| EC-12 | P1       | Planted/deceptive indicator                  | Independent corroboration + warning                                            | One-off planted indicator flagged                   |
| EC-13 | P1       | Competing hypotheses                         | Conflict sets and competing-link UI                                            | Both candidates are displayed                       |
| EC-14 | P0       | Wrongly accepted merge                       | Reversible/versioned memberships                                               | Rejection/split preserves history                   |
| EC-15 | P0       | Replayed content with dates changed          | Dedup score but preserve observation history                                   | Score remains stable                                |
| EC-16 | P0       | Forged/missing timestamps                    | Claimed/observed/captured times + confidence                                   | Timeline marks unreliable date                      |
| EC-17 | P1       | Clock skew                                   | UTC/NTP/server event time/sequence                                             | Audit ordering remains deterministic                |
| EC-18 | P0       | Topic/template drives MiniLM similarity      | Template stripping + score cap                                                 | Boilerplate pair remains weak                       |
| EC-19 | P0       | Short/multilingual/translated text           | Minimum corpus/language gates                                                  | No invalid style evidence                           |
| EC-20 | P1       | LLM imitation/style evasion                  | Support-only label + non-text corroboration                                    | Imitation fixture not high tier                     |
| EC-21 | P0       | Quote/signature/template contamination       | Cleaning pipeline/version/corpus hash                                          | Template excluded from features                     |
| EC-22 | P1       | Model/config version change                  | Versioned rescore/input hashes                                                 | Old/new link versions preserved                     |
| EC-23 | P0       | Missing category evidence misread as zero    | Category state labels                                                          | UI says not available/insufficient                  |
| EC-24 | P0       | Duplicate evidence score inflation           | Noisy-OR + groups                                                              | Duplicate test bounded                              |
| EC-25 | P1       | Tier fluctuates at boundary                  | Hysteresis + score-change event                                                | Small                                               |

variance does not flip tier |  
 | EC-26 | P0 | Unauthorized/unexplained decision | RBAC + mandatory reason + audit |  
 Decision without note rejected |  
 | EC-27 | P1 | Audit tampering | Append-only/tamper-evident strategy | Modification attempt  
 detected/logged |  
 | EC-28 | P0 | Sensitive evidence leaks | Redaction/access controls/masked logs |  
 Unauthorized/raw/export tests fail closed |  
 | EC-29 | P1 | Bulk insider enumeration | Rate limits/query audit/alerts | Repeated scans  
 limited/audited |  
 | EC-30 | P0 | Secret in Git/container/fixture | CI secret scan/env separation | Scan  
 passes; `.env` ignored |  
 | EC-31 | P1 | Compromised dependency | SBOM/lock/digest/vuln scan | Dependency report  
 available |  
 | EC-32 | P1 | Graph migration loses data | Adapter/dual-write/reconciliation/rollback |  
 Projection equals canonical records |  
 | EC-33 | P1 | Search stale/inconsistent | Canonical store first; index reconciliation |  
 Rebuild restores search |  
 | EC-34 | P1 | Concurrent processing | Idempotency/unique keys/locking | Two workers create  
 one canonical result |  
 | EC-35 | P0 | Export changes mid-build | Snapshot manifest/version lock | Report reproduces  
 original set |  
 | EC-36 | P1 | Retention/takedown/redaction | Retention/legal hold/tombstone policy |  
 Authorized redaction keeps audit notice |  
 | EC-37 | P0 | Air-gapped startup/download failure | Bundled artifacts/checksums/no runtime  
 download | Demo runs offline |  
 | EC-38 | P1 | Graph too large/cyclic/sensitive | Limits/progressive expansion/permission  
 filter | Truncated graph state displayed |  
 | EC-39 | P0 | Empty result misinterpreted | Explicit absence state | UI distinguishes  
 causes |  
 | EC-40 | P0 | Demo contains live/illegal content | Synthetic fixture policy/CI scan | No  
 live onion URL in artifacts |

## ## Required branch mapping

| Branch                  | Must close                                              |
|-------------------------|---------------------------------------------------------|
| 0 Baseline audit        | EC-30, EC-37, EC-40 inventory and current gaps          |
| 1 Evidence contract     | EC-10, EC-11, EC-14, EC-16, EC-23                       |
| 2 Collection/provenance | EC-01, EC-02, EC-04, EC-15, EC-16, EC-28, EC-34 partial |
| 3 Fusion/lifecycle      | EC-08, EC-09, EC-11, EC-13, EC-14, EC-23, EC-24, EC-25  |
| 4 OnionScan             | EC-07, EC-08, EC-31 partial                             |
| 5 Dashboard/review      | EC-13, EC-26, EC-28, EC-39                              |
| 6 Timeline              | EC-16, EC-17, EC-39                                     |
| 7 Behavior/migration    | EC-12, EC-18, EC-20, EC-21                              |
| 8 Classical style       | EC-18, EC-19, EC-20, EC-21, EC-22                       |
| 9 Export/governance     | EC-05, EC-26 through EC-36                              |
| 10 Graph migration      | EC-32, EC-33, EC-38                                     |
| 11 Demo hardening       | EC-01, EC-07, EC-37, EC-40 regression tests             |

## ## Stop-the-line criteria

Do not demo live collection or proceed to pilot when any of the following is true:

- A visible graph edge lacks an evidence ID or caveat.
- The system can claim a real-world identity.
- Direct collector egress bypasses Tor/proxy policy.
- Raw sensitive content is exposed to an unauthorized role.
- A score can be inflated by duplicate/reposted evidence.
- Text-only signals can yield Likely Same Actor or Observed Technical Identity.
- Export cannot reproduce exact evidence/link/model versions.
- Demo fixtures include live onion URLs, real credentials or illegal content.